

ESP32 7-Segment Display Control using Blynk IoT

Objective

Control a single 7-segment display connected to an ESP32 through the Blynk IoT platform. The selected digit (0–9) from the Blynk slider is displayed on the 7-segment display in real time.

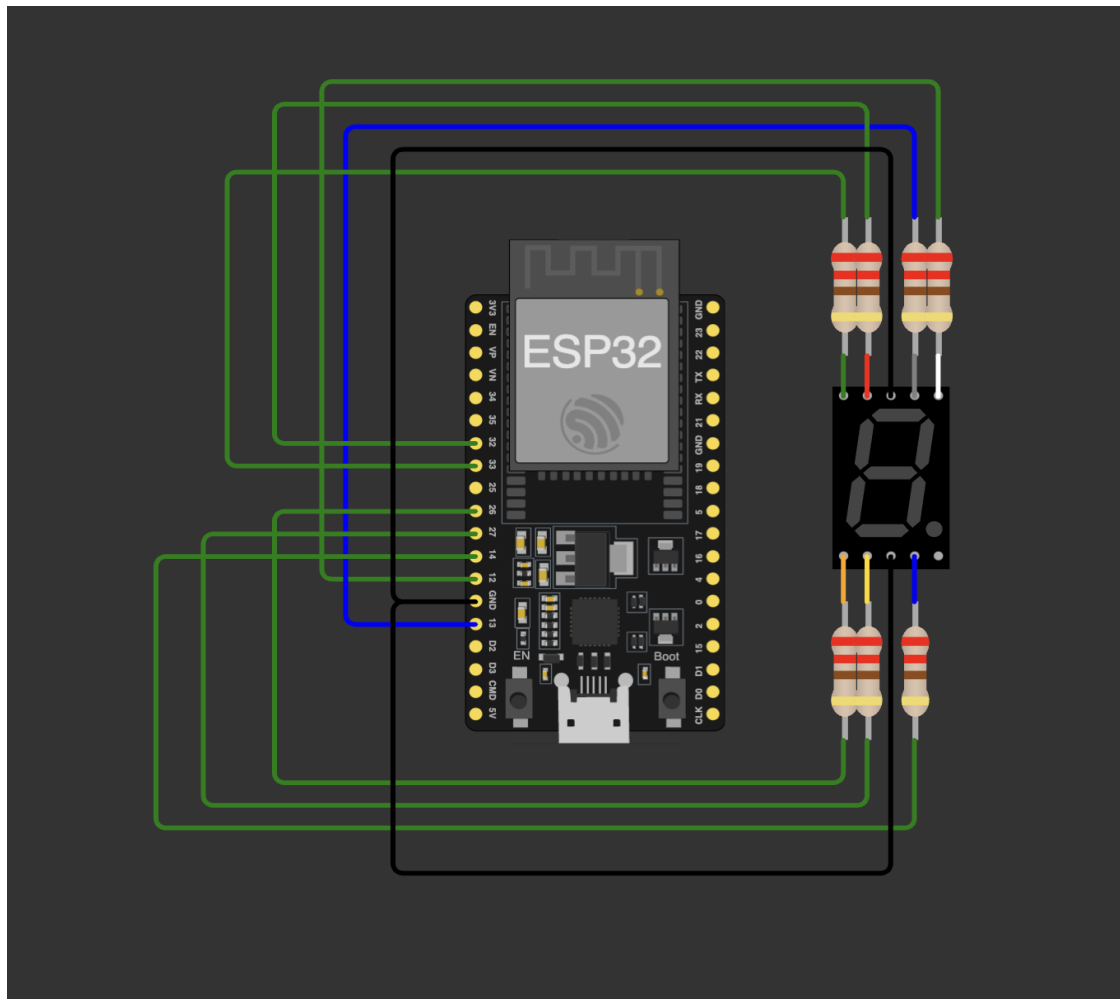
Components Required

ESP32 DevKit, Single 7-Segment Display, 220Ω Resistors, Jumper Wires, Wi-Fi Connection and Blynk IoT Platform.

Working Principle

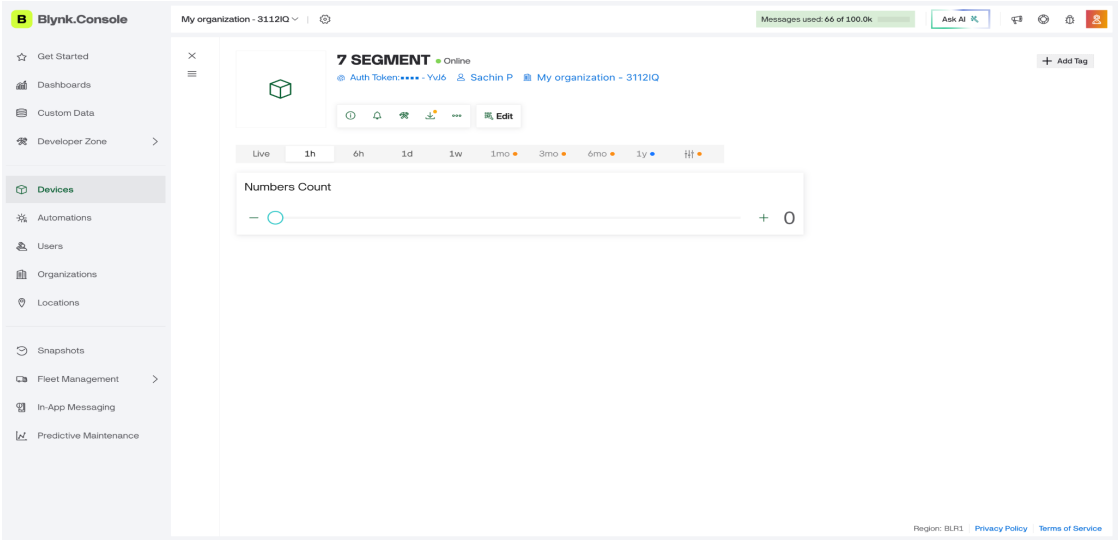
The Blynk slider (Virtual Pin V0) sends a value from 0 to 9. ESP32 receives the value using `BLINK_WRITE(V0)` and lights the corresponding segments to display the selected digit.

Circuit Diagram

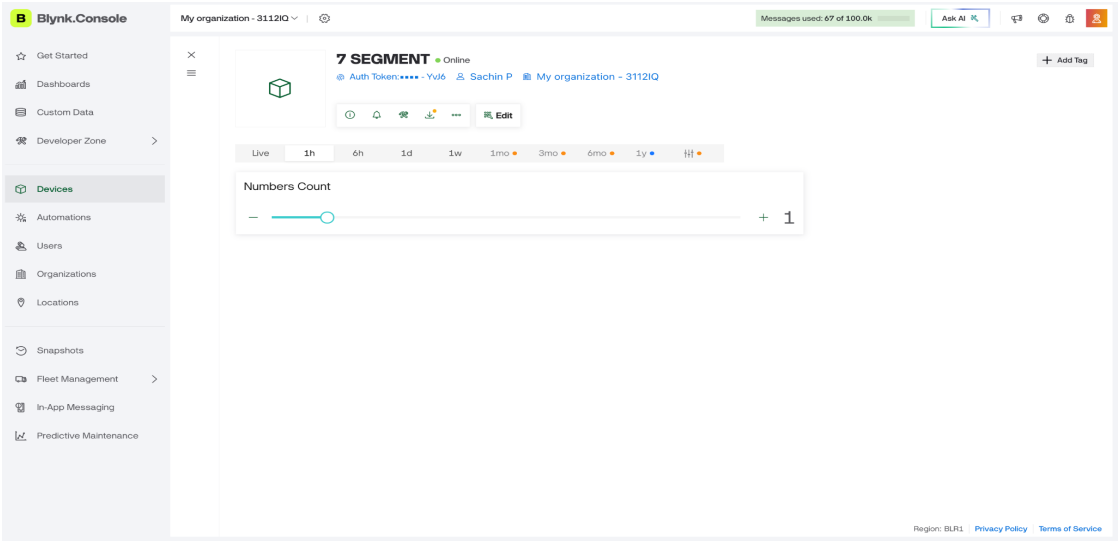


Blynk Dashboard Inputs (0–9)

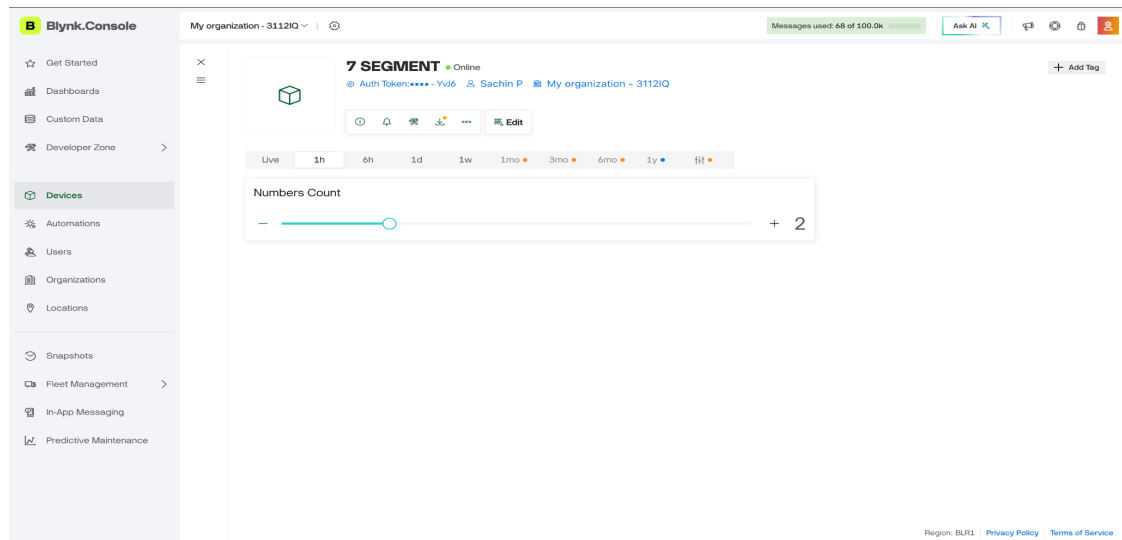
Input 0



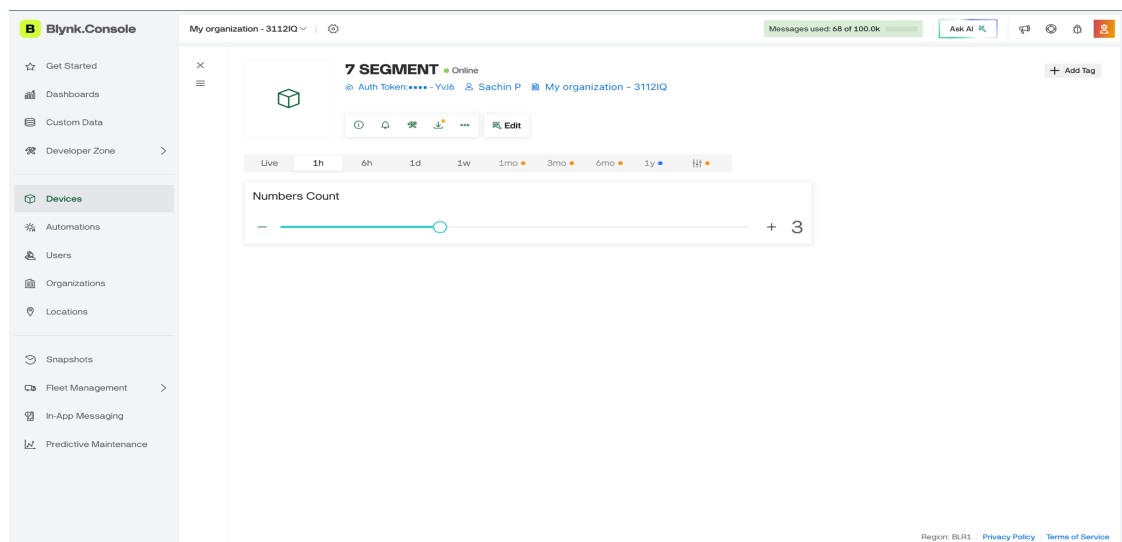
Input 1



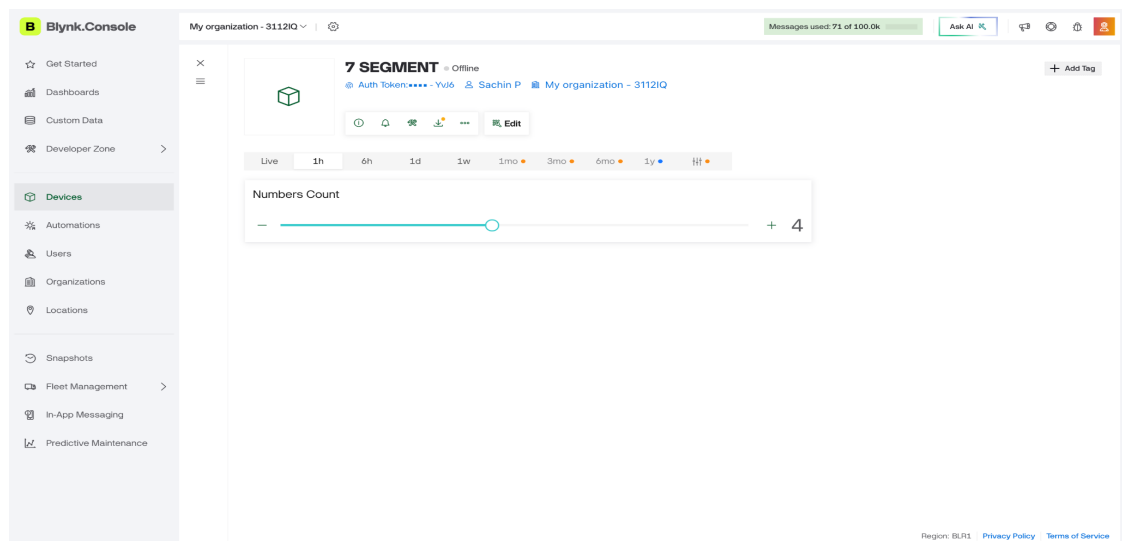
Input 2



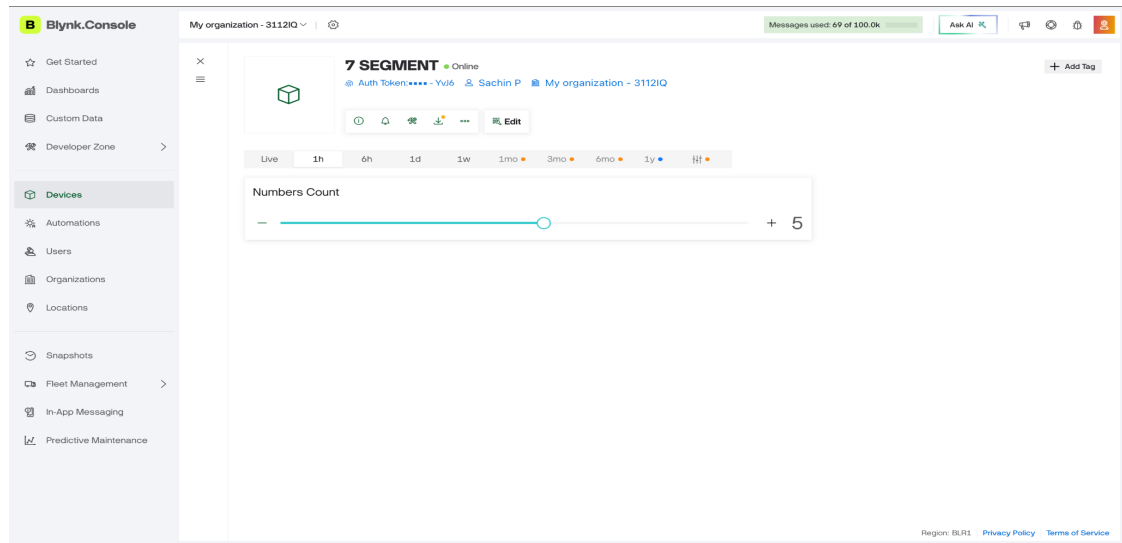
Input 3



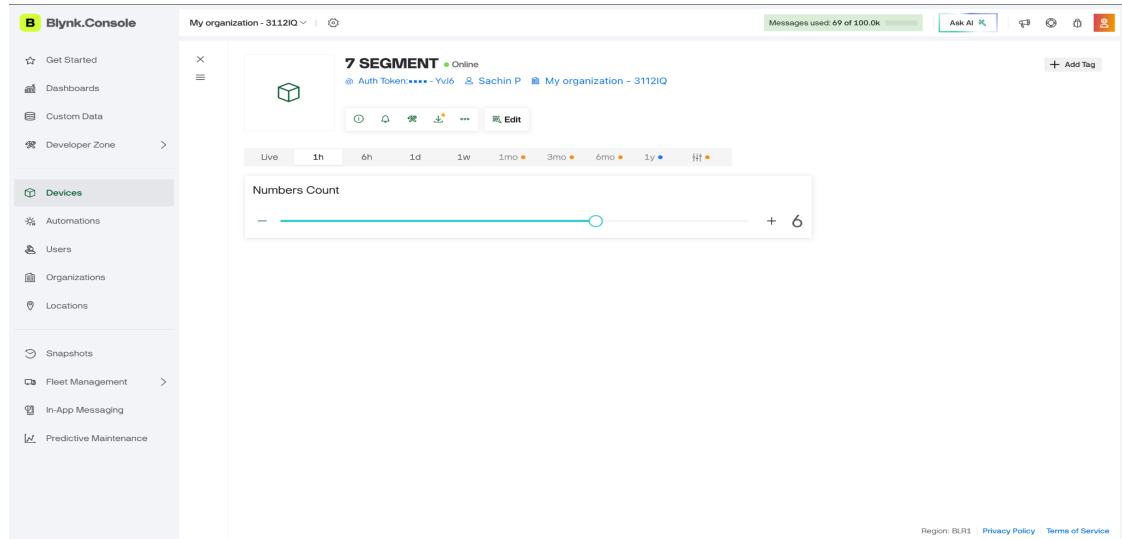
Input 4



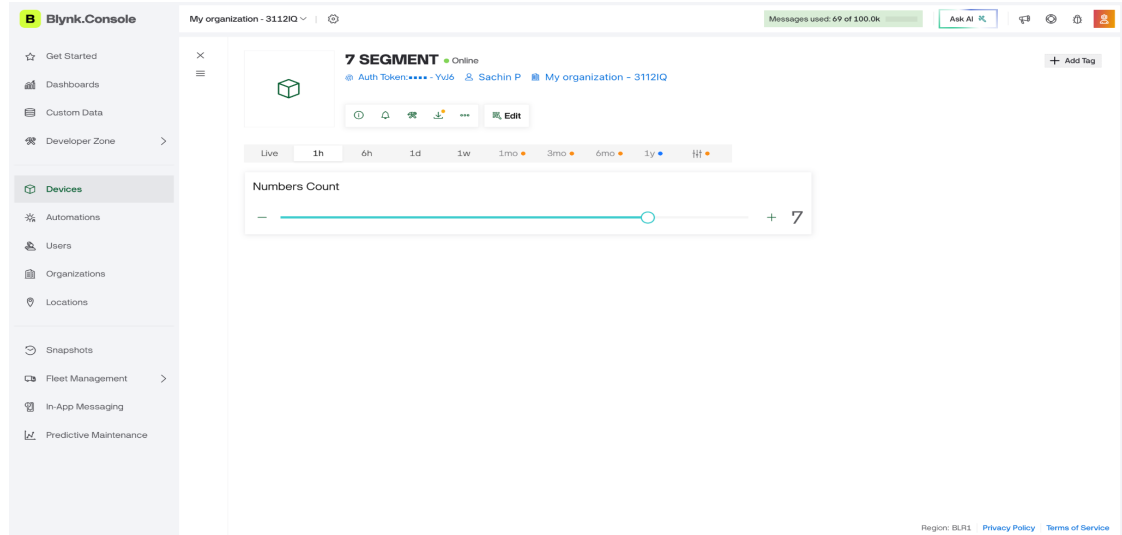
Input 5



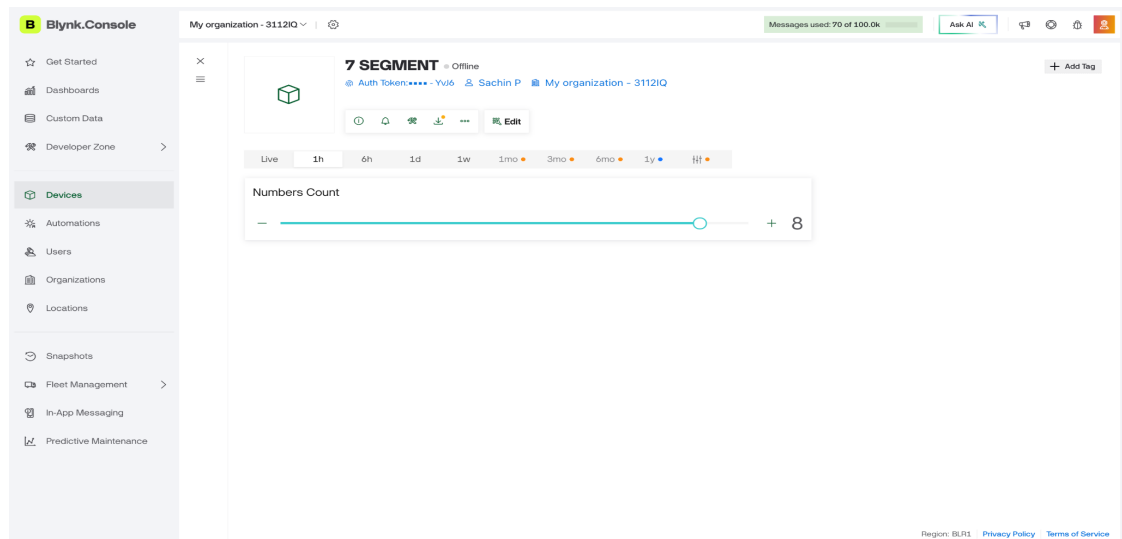
Input 6



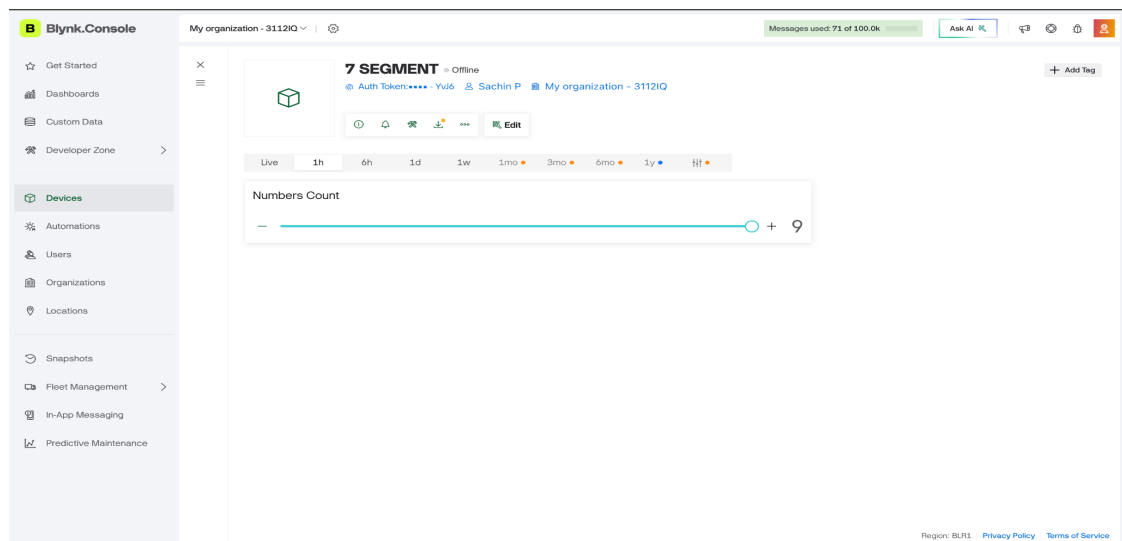
Input 7



Input 8

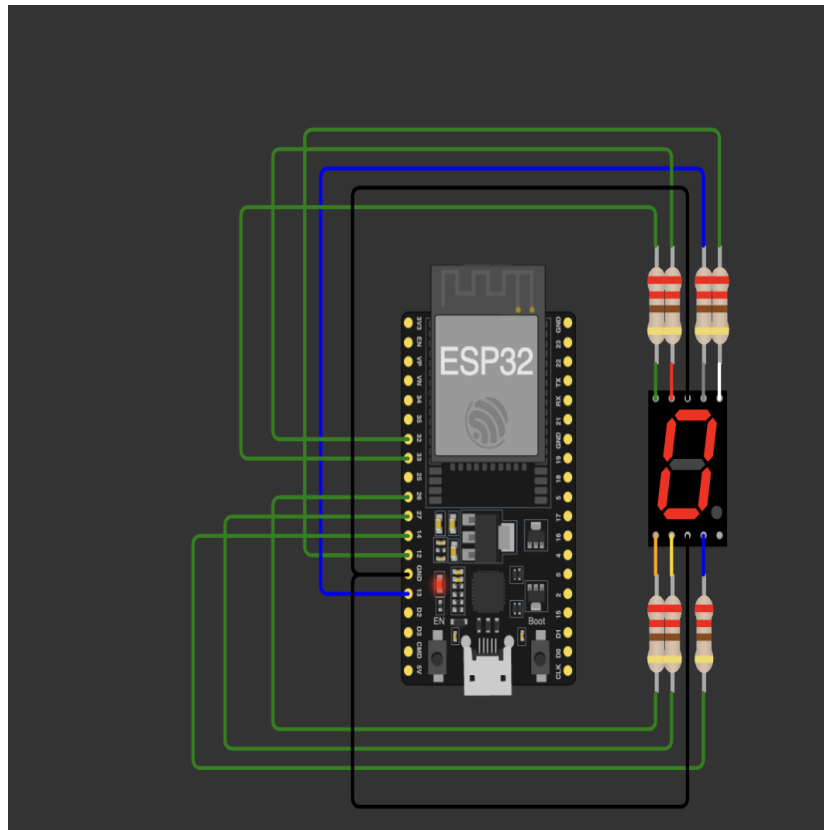


Input 9

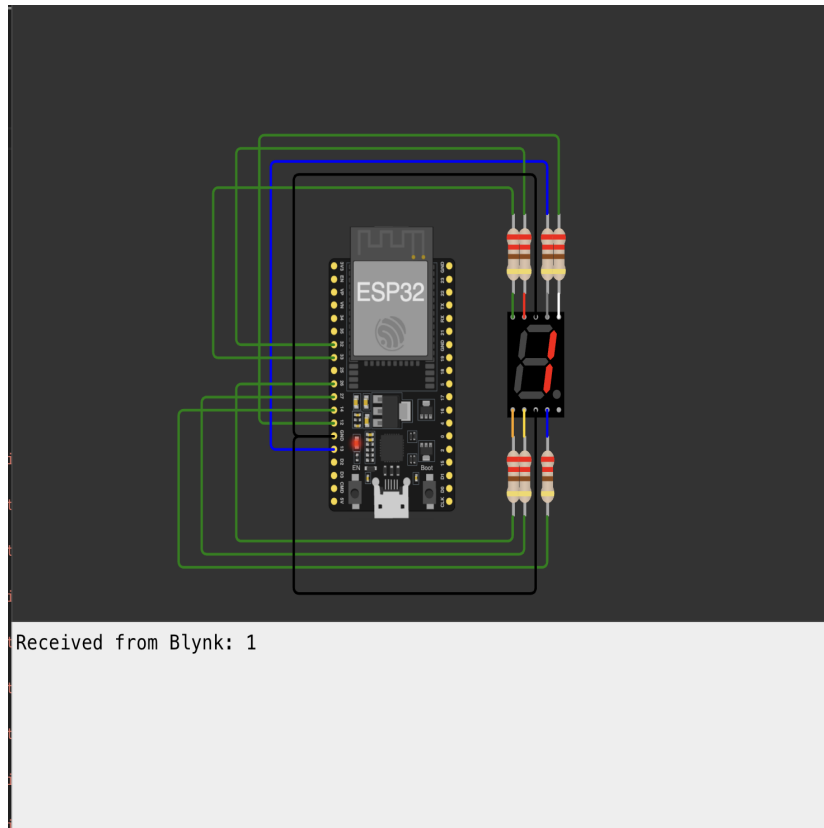


Simulation Outputs (0–9)

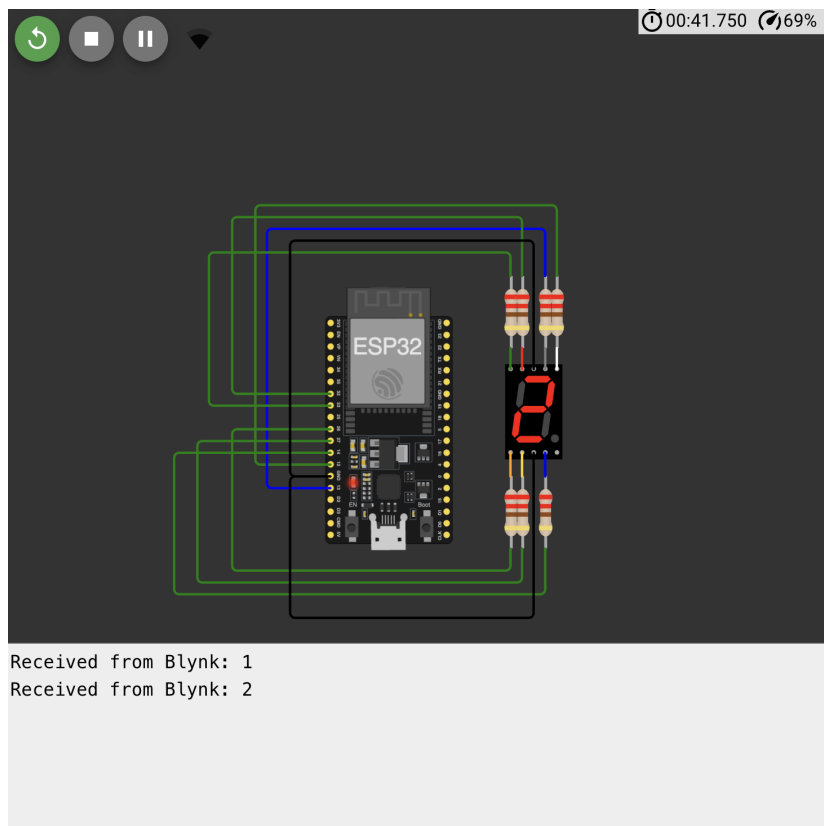
Output 0



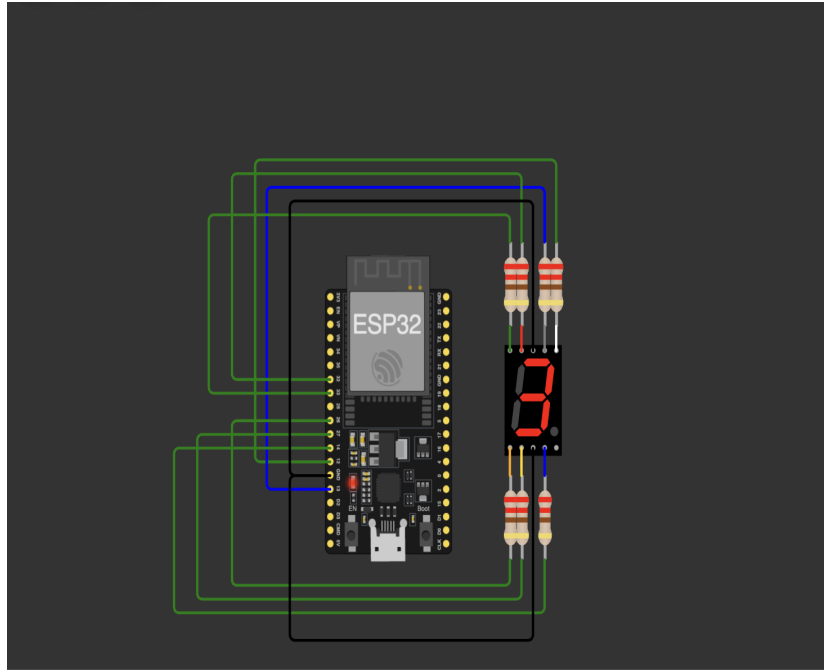
Output 1



Output 2

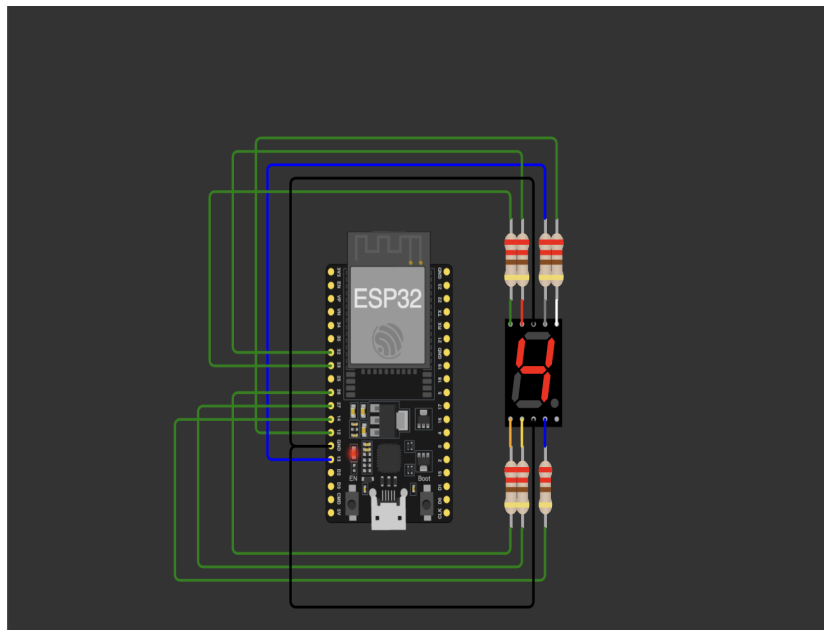


Output 3



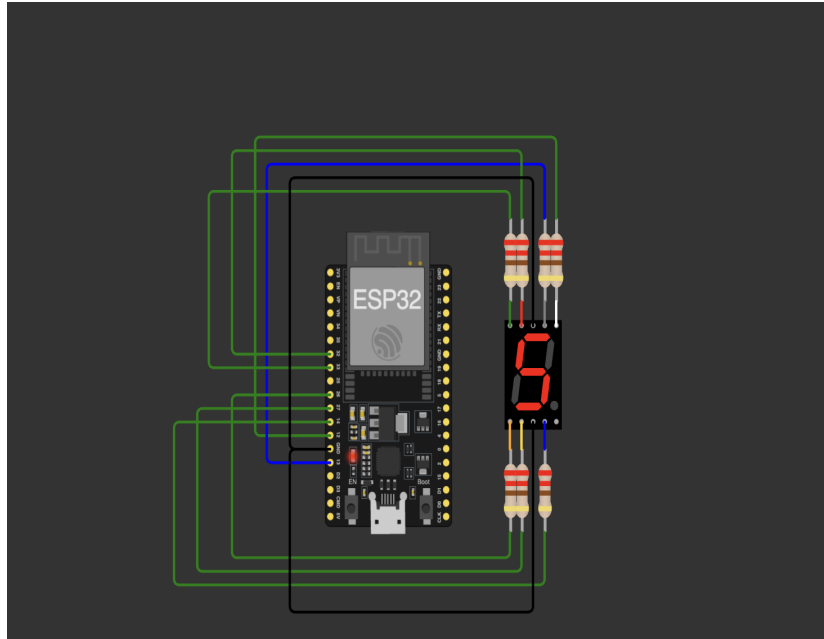
Received from Blynk: 1
Received from Blynk: 2
Received from Blynk: 3

Output 4



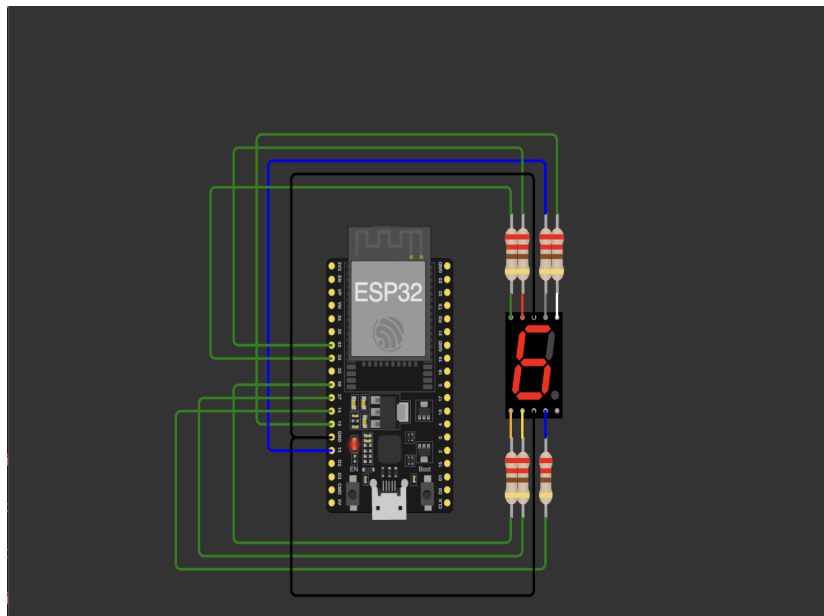
Received from Blynk: 1
Received from Blynk: 2
Received from Blynk: 3
Received from Blynk: 4

Output 5



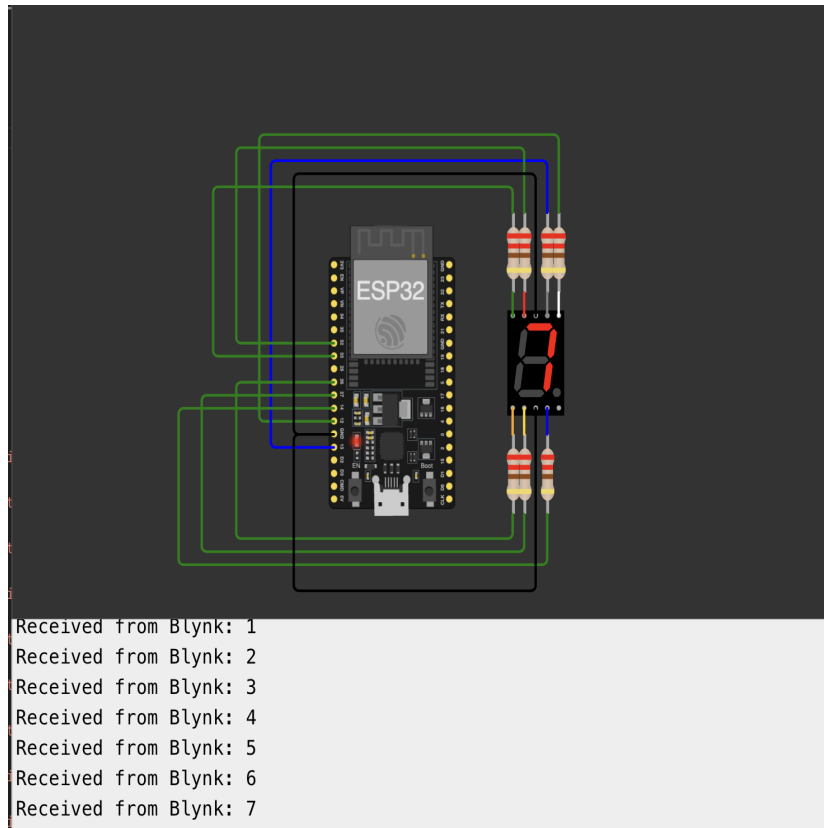
Received from Blynk: 1
Received from Blynk: 2
Received from Blynk: 3
Received from Blynk: 4
Received from Blynk: 5

Output 6

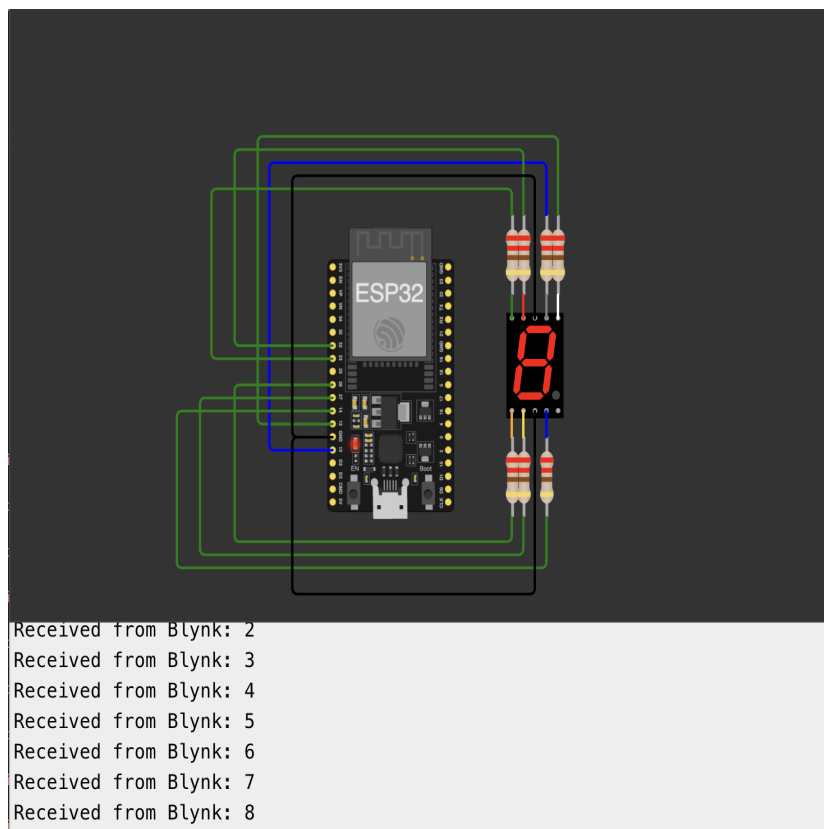


Received from Blynk: 1
Received from Blynk: 2
Received from Blynk: 3
Received from Blynk: 4
Received from Blynk: 5
Received from Blynk: 6

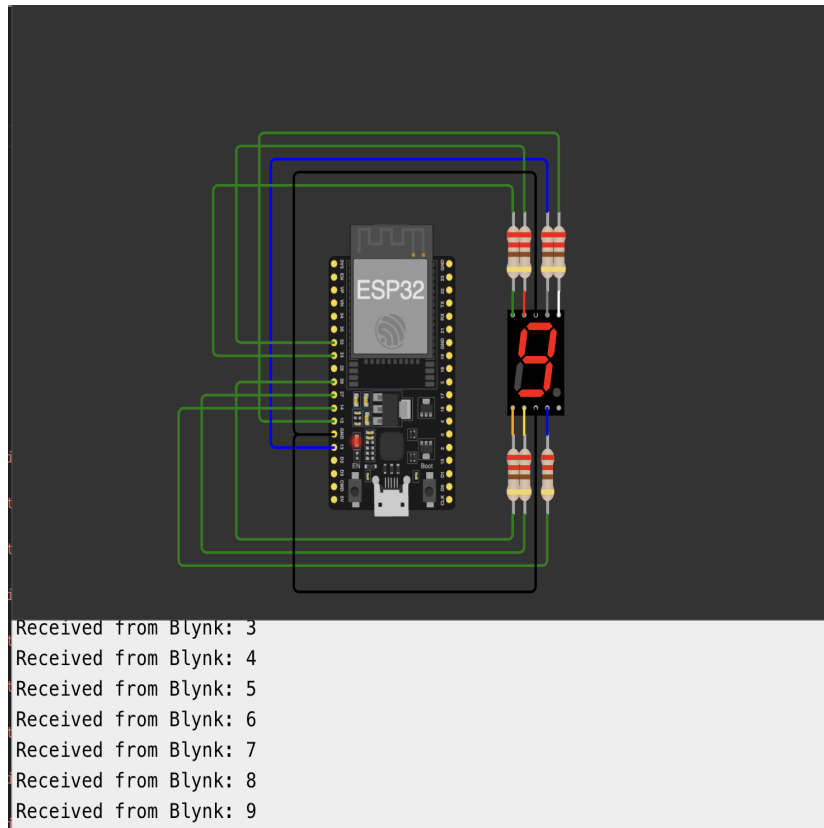
Output 7



Output 8



Output 9



Conclusion

The ESP32 successfully receives values from the Blynk slider and displays the corresponding digit from 0 to 9 on the 7-segment display. Real-time communication between the Blynk IoT platform and ESP32 was verified through the simulation outputs and Serial Monitor.